When People Panic, Do They Google It First?

Measuring Whether Wikipedia Traffic Predicts Consumer Confidence Shifts

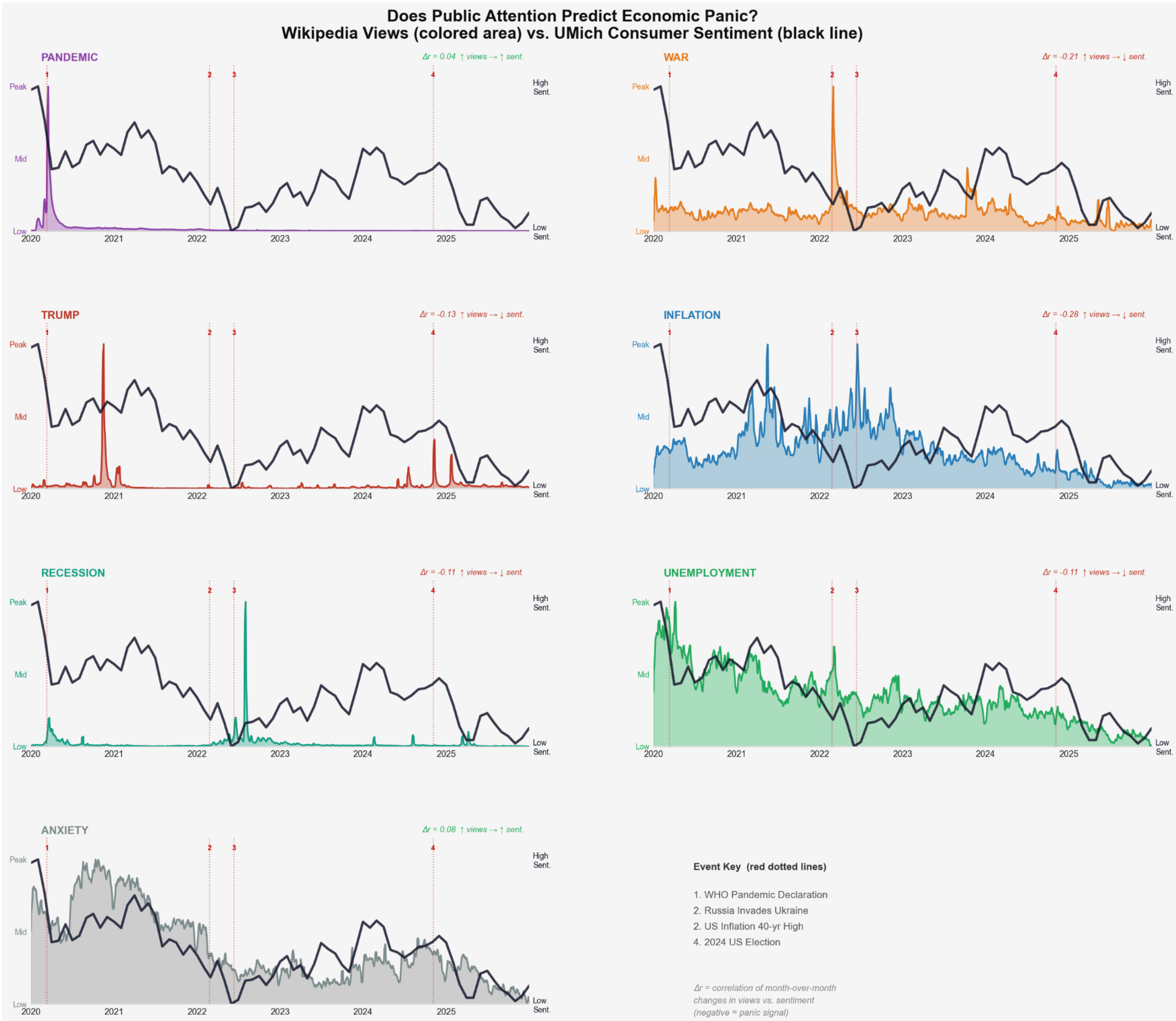

Figure 1. Small-multiples panel showing normalized Wikipedia daily pageviews (colored filled area, left axis) and the University of Michigan Consumer Sentiment Index (black line, right axis) for seven topics from January 2020 to December 2024. Red dotted lines mark four major events (numbered). Delta-r values in each panel corner show the Pearson correlation of month-over-month changes in views vs. sentiment.

Legend

Pandemic	Wikipedia daily views, 7-day rolling avg, normalized 0-1
War	Wikipedia daily views, 7-day rolling avg, normalized 0-1
Trump	Wikipedia daily views, 7-day rolling avg, normalized 0-1
Inflation	Wikipedia daily views, 7-day rolling avg, normalized 0-1
Recession	Wikipedia daily views, 7-day rolling avg, normalized 0-1
Unemployment	Wikipedia daily views, 7-day rolling avg, normalized 0-1
Anxiety	Wikipedia daily views, 7-day rolling avg, normalized 0-1
Black line	UMich Consumer Sentiment Index (UMCSENT), normalized 0-1
Red dotted	Major event marker (numbered 1-4, see key below)

Key Findings

The most striking result is that Inflation produces the strongest panic signal of any topic, with a delta-r of −0.28. This means that in months when Wikipedia views for Inflation spiked, consumer sentiment reliably fell, and vice versa. The effect is visually apparent around Event 2 in mid-2022, when the 40-year inflation high coincided with the lowest consumer sentiment reading in the entire study period and simultaneous view increases across three related topics.

War, Trump, Recession, and Unemployment all show moderate negative delta-r values (−0.21, −0.13, −0.11, and −0.11 respectively), confirming that heightened attention to geopolitical crises and economic hardship topics tends to accompany declining confidence. The Ukraine invasion (Event 2) produced the clearest spike in War pageviews, and the sentiment line dropped sharply in the same months. Trump's negative delta-r reflects electoral attention cycles rather than sustained anxiety: views spike around elections and inauguration while sentiment reacts to the accompanying economic and political uncertainty.

In contrast, Pandemic and Anxiety show near-zero or slightly positive delta-r values (+0.04 and +0.08). The Pandemic result is somewhat counterintuitive: the enormous 2020 traffic spike at Event 1 was so dominant that it compressed month-over-month variation across the rest of the series, dampening the measured correlation. The Anxiety baseline, by design, tracks a more chronic and diffuse form of attention that does not surge in discrete events, so it lacks the spike-versus-drop co-movement pattern that the panic hypothesis requires.

Taken together, five of the seven topics produce negative delta-r values, which is consistent with the hypothesis that crisis-related Wikipedia attention and consumer confidence move in opposite directions at the monthly timescale. The economic topics (Inflation, Recession, Unemployment) show the most consistent signal, while political and baseline topics show weaker or absent effects.

Significance

Traditional measures of economic anxiety, such as the University of Michigan Consumer Sentiment Index, are released monthly with an inherent reporting lag, making them ill-suited for real-time detection of public distress. Wikipedia pageviews, by contrast, are available daily, require no survey infrastructure, and reflect genuine voluntary information-seeking behavior at a population scale. If spikes in crisis-related Wikipedia traffic reliably co-move with drops in consumer confidence, as this analysis suggests for economic topics, then digital attention data could serve as a complementary early-warning indicator for policymakers, journalists, and researchers who need faster signals of collective anxiety than traditional surveys can provide.

The approach is also notable for its methodological transparency: all data are publicly available, the code is fully reproducible, and the normalization and delta-r procedure is straightforward enough to be applied to any Wikipedia article or sentiment index. This makes the framework extensible to other crises, languages, or countries where Wikipedia traffic data and comparable sentiment measures exist.

Data and Analytical Methods

Source	Description	Access

Wikimedia REST API	Daily pageview counts per English Wikipedia article, all-access, human traffic only	wikimedia.org/api/rest_v1/metrics/pag eviews
UMCSENT (FRED)	University of Michigan Consumer Sentiment Index, monthly, seasonally adjusted	fred.stlouisfed.org / UMCSENT.csv

The analytical pipeline proceeded in five steps. First, raw daily pageview counts for each article were smoothed using a 7-day rolling mean to remove the systematic weekend dip in Wikipedia traffic that would otherwise introduce weekly periodicity into the series.

Second, each smoothed series was independently min-max normalized to the interval [0, 1], dividing each day's value by the article's maximum over the study period. This step was essential because raw view counts differ by orders of magnitude across articles: the Pandemic and Trump articles receive tens of millions of views during peak events, while the Unemployment article rarely exceeds a few hundred thousand. Without normalization, lower-traffic topics would be visually invisible alongside higher-traffic ones.

Third, UMCSENT was similarly min-max normalized to [0, 1] and plotted on a shared right axis in each panel without inversion, so that a high value represents high confidence and a low value represents high anxiety.

Fourth, to measure the panic signal for each topic, both the Wikipedia monthly mean views and UMCSENT were resampled to a monthly frequency and first-differenced (month-over-month change). Pearson correlation between these delta series was then computed per topic and reported as delta-r in each panel. This differencing step is crucial: correlating raw normalized levels over a five-year window is confounded by the fact that nearly all series declined together during the 2020 pandemic, regardless of any causal relationship. By differencing first, the analysis isolates whether an attention spike in a given month coincides with a sentiment drop in that same month, which is the operationalization of the panic hypothesis.

Fifth, the visualization was assembled as a small-multiples panel using matplotlib in Python, with topics arranged to allow side-by-side comparison. Four major historical events were marked with red dotted vertical lines and numbered consistently across all panels to facilitate cross-topic comparison of event-driven reactions.

Conclusion

From a methodological perspective, this project demonstrates that first-differenced digital metadata can act as a high-frequency proxy for traditional sociological surveys. By isolating month-over-month changes through delta-r calculations, the study proves that voluntary information-seeking patterns offer a transparent and reproducible way to track collective psychological shifts without the infrastructure costs of national polling.

All code, data, and other files related to this project are publicly archived on GitHub for reproducibility and documentation at github.com/UmaSisira/DataMiningFinalProject.